

Initial Project

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Course: Computer Architecture Essentials Assessments

Book: Initial Project

Printed by: Noor Dyer

Date: Wednesday, 8 April 2026, 6:02 PM

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Project

Faculty:	Information Technology
Module Code:	ITCAA2-33
Module Name:	Computer Architecture Essentials
Content Writer:	Arsene Mpiana
Internal Moderation:	Community of Practice
Copy Editor:	Kyle Keens
Total Marks:	100
Submission Week:	Week 7

This module is presented on NQF level 6.

5% will be deducted from the student's assignment mark for each calendar day the assignment is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Projects submitted later than three calendar days after the deadline or not submitted will get 0%. ^[1]

This is a group project.

Group consist of 4 - 6 members

This project contributes 50% towards the final mark.

[1] Under no circumstances will assignments be accepted for marking after the assignments of other students have been marked and returned to the students.

Instructions to Students

1. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
2. Remember to keep a copy of all submitted projects.
3. All work must be typed.
4. Please note that you will be evaluated on your writing skills in all your projects.
5. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
6. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.
7. One group member should be nominated to submit the assessment on behalf of the group. Multiple submissions by various group members will result in an inflated similarity index on Turnitin.

Section A

Learning Objective

Understand the architecture and hardware components of the Raspberry Pi.

Develop low-level programs using assembly and C/C++ languages.

Integrate Raspberry Pi with external devices using GPIO pins.

Explore advanced features such as multithreaded programming, sound integration, and graphical programming.

Project Topic

Smart Irrigation

Scope

Week 1 - 7

Technical Aspects

Arduino

Question 1

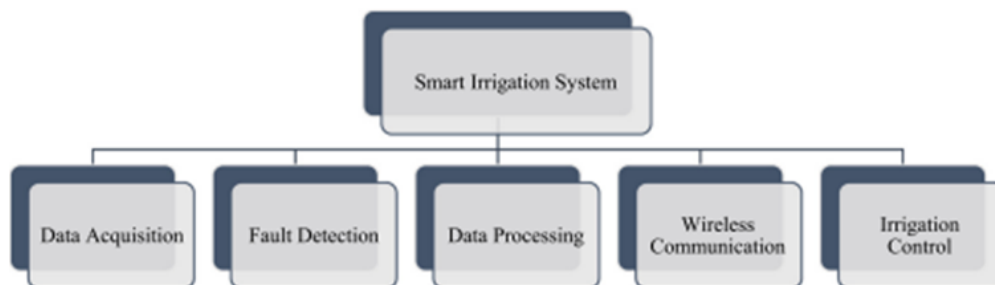
20 Marks

Study the scenario and complete the question(s) that follow(s):

Smart Irrigation

Agriculture is an important industry as well as the foundation of the economy. Agriculture automation is a major concern and emerging topic for all countries. The world's population is rapidly increasing, and as the population grows, the demand for food increases. The developing need for food, as well as changing consumer demands, have made it extremely difficult for the agriculture industry to develop techniques and practices that will allow them to fully satisfy the increasing needs and requirements. Smart irrigation is emerging as new scientific disciplines that use data-intensive methods to increase agricultural productivity while reducing its environmental impact. Modern agricultural operations generate data from a variety of sensors, leading to a better understanding of both the operational environment and the operational activities. This enables more accurate and efficient decision-making. In addition, this allows for optimising resources and achieving the intended objectives from this sector. The water is conserved when implementing these technologies in the irrigation system.

The block diagram of a smart irrigation is given below



Conduct a critical study to analyse five benefits of implementing and deploying an IoT monitoring system in irrigation.

(10 Marks)

End of Question 1

Question 2

75 Marks

Study the scenario and complete the question(s) that follow(s):

Implementing Smart Irrigation

IoT smart irrigation systems incorporate various sensors, such as soil moisture sensors, weather sensors, humidity sensors, and temperature sensors. These sensors continuously monitor environmental conditions and soil moisture levels in real time.

IoT smart irrigation systems are typically connected to the internet, allowing the sensors and controller to transmit data and receive commands from cloud-based platforms. This enables the user to remotely monitor and control the irrigation system from anywhere with an internet connection.

Project Specifications

You are tasked to design an irrigation system controlled by microprocessor-based circuitry capable of fully automatic, semi-automatic, and manual control.

Design the Irrigation System with the following characteristics:

2.1 The irrigation system should have three sensors to detect if an area is dry or not.

(6 Marks)

2.2 The irrigation system should have three sprinkler heads, each located close to the sensors described above and each sprinkler head should open automatically if a sensor has detected a dry area next to it.

(10 Marks)

2.3 A servo motor is used to indicate the operation of the pump and should have three states operations indicating the required speed of the pump based on the number of areas being watered. A display should be used to indicate the area being watered.

(10 Marks)

2.4 The Watering period should be controlled by a switch and should be set to a maximum of 30 seconds. While watering an area, if the switch is closed the watering process should end otherwise, it should ends after 30 seconds.

(10 Marks)

2.5 Once the watering process has started, the remaining number of seconds of the watering period should be displayed.

(6 Marks)

2.6 It is important to ensure that a person or animal should not enter an area during watering period as such a sensor should be used to detect an intrusion in the area being watered. An alert action is required to prevent intrusion should be performed in case of a detection.

(10 Marks)

2.7 The irrigation system should operate at high performance by implementing multithreading of tasks.

(10 Marks)

2.8 After 30 times that the irrigation system has watered, maintenance is required, in this case a maintenance sign should be displayed.

End of Question 2

Question 3

20 Marks

Study the scenario and complete the question(s) that follow(s):

Implementing Smart Irrigation

A smart irrigation system can also be set up to send alerts or notifications to users' smartphones or other devices when adjustments are made. An irrigation controller is a device to operate automatic irrigation systems such as lawn sprinklers and drip irrigation systems. It is important to write a report of your project because it indicates the work performed in the development of your system and how it can be used to ensure smart irrigation.

You are required to write a report on your project that includes the following:

3.1 Draw the block diagram of your project showing different main part of the system.

(5 Marks)

3.2 Draw the circuit diagram of your project showing different components of your system.

(5 Marks)

3.3 Show the code of your developed system.

(5 Marks)

3.4 By the mean of images and snippet, perform the simulation of your system to show the operation and discuss the result.

(5 Marks)

End of Question 3

Appendix

REPORT FORMAT

Title/Cover Page

Abstract

Table of Contents

1. Introduction

1.1 Background: Brief introduction of the importance of agriculture and the role of smart irrigation in addressing modern challenges (using the given scenario as a guide).

1.2 Problem Statement

1.3 Project Objectives: Outlines the main goals of the smart irrigation system.

1.4 Critical Analysis of IoT Benefits in Irrigation (Addressing Q1.1): Present and critically analyze five significant benefits of implementing IoT monitoring systems in irrigation (e.g., water conservation, optimized plant health, cost reduction, remote management, data-driven decision making, environmental sustainability). For each benefit, explain how IoT enables it and its impact.

2. System Requirements and Specifications

2.1 Functional Requirements: List the core functionalities the system must perform (e.g., automatic watering, intrusion detection).

2.2 Non-Functional Requirements: Mention aspects like performance, reliability, user-friendliness.

2.3 Detailed Project Specifications (Summarizing Q2.1 - Q2.8): Reiterate the specific design characteristics required for the system (e.g., 3 sensors, 3 sprinklers, servo pump control, watering timer, intrusion detection, multithreading, maintenance indicator).

3. System Design

3.1 High-level Block Diagram (Addressing Q3.1):

Present and explain the high-level block diagram of the smart irrigation system, showing the main functional blocks and their interconnections.

3.2 Hardware Design:

3.2.1 Component Selection: Justify the choice of microcontroller (Raspberry Pi Pico/Arduino), sensors (soil moisture, PIR/ultrasonic), actuators (sprinkler valves/relays, servo motor), display (LCD/OLED), and switches.

3.2.2 Circuit Diagram (Addressing Q3.2): Provide a detailed circuit diagram, clearly labeling all components and connections.

3.3 Software Design:

3.3.1 Flowchart/State Diagram: Illustrate the overall program flow or the different operational states of the system (automatic, semi-automatic, manual).

3.3.2 Multithreading Implementation (Addressing Q2.7): Discuss how multithreading is implemented in the code. Identify the specific tasks that are multithreaded and explain why this approach was chosen to achieve high performance.

4. Software Implementation

4.1 Developed Code (Addressing Q3.3): Include the complete source code for the smart irrigation system.

4.2 Development Environment: Brief description of the tools used (IDE, compiler, libraries).

5. Testing, Results, and Discussion

5.1 Simulation/Testing Methodology (Addressing Q3.4): Describe how the system was tested.

5.2 Results (Addressing Q3.4 & Q2.1-2.8): For each of the project specifications (Q2.1-Q2.8), student should provide images (screenshots from simulation or photos of physical setup) and relevant code snippets to demonstrate its successful operation. Discuss the results for each feature, confirming how the system meets the requirement. Specifically, for Q2.7 (Multithreading): Provide evidence or a discussion of how multithreading improves performance

5.3 Challenges and Solutions: Discuss any problems encountered during design or implementation and how they were overcome.

5.4 Performance Evaluation: Critically evaluate the system's performance against the initial requirements.

6. Conclusion

Summary of the implemented system and discussion of potential future enhancements or developments for the system.

7. References

Appendices (Optional)

- Full code (if too long for main body)
- Detailed component datasheets
- Additional images/diagrams